

PRATHMESH DESHKAR

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SKILLS

Languages & Frontend: Java, TypeScript, JavaScript, HTML/CSS, React, Vite, Tailwind CSS, Radix UI
Backend & Database: Java 21, REST API, JDBC, Gradle, JWT, MongoDB Atlas, GridFS, MySQL, phpMyAdmin
DevOps & Tools: Docker, Render, Electron, GitHub Actions, VS Code, Git, Linux, Kali Linux
Core CS: Data Structures, Algorithms, OOP, Time & Space Complexity

EXPERIENCE

Software Development Research – Java (Independent Project) Jan 2023 – Dec 2023

- Developed and implemented a custom enhancement of the **AES-192** encryption algorithm in Java.
- Conducted quantitative analysis using **entropy**, **avalanche effect**, and **execution time** to **evaluate security-performance trade-offs** between security and performance.
- Applied **OOP principles**, **software testing**, and **cryptographic evaluation**, Published results in **IJSRD**, demonstrating applied **research** and **secure software design**.

PROJECTS

Sentra - Encrypted File Sharing Platform [Github[^]] [Live Link[^]] Jan 2026 – Apr 2026

- Full-stack encrypted file-sharing platform: **React + TypeScript** frontend, **Java 21 backend**, deployed on **Render**.
- AES-256-GCM** encryption, **RSA** key wrapping, **JWT auth** with HttpOnly cookies, refresh tokens, and rate limiting.
- MongoDB Atlas + GridFS** for encrypted storage; inbox/outbox lifecycle with file expiry and self-destruct.
- Dual-platform release: web app (**Vite + Tailwind**) and **Electron**-based Windows desktop with auto-update.
- Technologies:** Java 21, React, TypeScript, Vite, Tailwind CSS, MongoDB Atlas, GridFS, AES-256-GCM, RSA, JWT, Electron, Docker, Render

Text Encryption using Modified AES [GitHub[^]] Jan 2023 – Dec 2023

- Built a **secure encryption tool** using a **modified AES-192 algorithm** proposed in my research paper.
- Integrated cryptographic evaluation metrics to **assess performance and security**.
- Technologies:** Java, AES, VS Code, Cryptography

EDUCATION

Madhav Institute of Technology and Science Gwalior, Madhya Pradesh
Bachelor of Technology – Information Technology Expected: Jun 2027
CGPA – 7.94 / 10 (as of 5th Semester)

Government Polytechnic College Betul, Madhya Pradesh
Diploma in Computer Science Completed: 2024
CGPA – 8.09 / 10

Secondary School Certificate (10th) Completed: 2021
CGPA – 8.7 / 10

ACHIEVEMENTS

- Published peer-reviewed research paper: "**Assessing the Impact of Increased MixColumns on AES Encryption Security and Performance**" in **IJSRD (International Journal of Scientific Research and Development)**.

CERTIFICATIONS & TRAININGS

- Software Engineering Virtual Simulation – Walmart (2025)**
Data structures, software architecture, relational databases, and data munging.
- Solutions Architecture Virtual Simulation – AWS (2025)**
Designed a scalable hosting architecture on Elastic Beanstalk; documented solution in plain language covering functionality and cost breakdown.
- Ethical Hacking – IIT Jodhpur (2022)**
Penetration testing, cryptography, and web security using Kali Linux, Metasploit, Nmap, and Burp Suite.